Automated Hallucination Detection & Evidentiary Verification Pipeline

An Open-Source Multi-Agent NLP Framework with Federated Dual-Stance Retrieval & DeBERTa Cross-Encoder Adjudication

Domain: Natural Language Processing / Fact Verification | **Framework:** PyTorch, DeBERTa-v3, FastAPI | **Execution:** 100% Free/Local

Executive Summary: *Large Language Models (LLMs) frequently generate convincing but factually incorrect assertions ('hallucinations'), inverted causal assertions, and ungrounded claims. This project delivers an end-to-end, deterministic, multi-agent verification system that ingests unstructured LLM responses, decomposes them into atomic claims, gathers dual-perspective evidence via federated search (PubMed SciFact, FEVER, & Live Web), and adjudicates factual veracity using cross-encoder Natural Language Inference (NLI) with redundancy-discounted effective sample size calibration.*

1. Problem Statement & Architectural Innovations

Standard verification approaches suffer from three critical flaws: (1) *Confirmation Bias* (retrieving only supporting evidence), (2) *Citation Double-Counting* (treating republished wire articles as independent proof), and (3) *Expensive API Latency* (relying on commercial LLM self-checks). Our architecture overcomes these limitations through four core innovations:

- **Strict Atomic Propositional Decomposition:** Preserves temporal, numerical, and spatial qualifiers so compound assertions cannot masquerade as universally valid.
- **Dual-Perspective Evidentiary Debate:** Operates an adversarial protocol between a Defender Agent (seeking corroboration) and a Prosecutor Agent (formulating targeted counter-queries).
- **Quality Firewall & Redundancy Clustering:** Clusters similar evidence and calculates Kish Effective Sample Size (n_{eff}) to discount redundant sources.
- **DeBERTa-v3 Cross-Encoder Adjudication:** Evaluates premise-hypothesis cross-attention without free-text hallucinations, yielding a calibrated Global Hallucination Index $H \in [0.0, 1.0]$.

2. Modular Pipeline Architecture (12 Discrete Stages)

Stage / Module	Core Function & Implementation	Output Artifact / Metric
M1: Intake & M2: Gen	Ingests user prompt, configures session tracking, accepts text input.	Audit Session Record
M3: Atomic Extraction	Rule-based NLP & regex clause splitter; extracts temporal/numeric/spatial anchors.	List of Claim objects (0..N)
M4: Complexity Analysis	Evaluates lexical & entity density; formulates verification priority plan.	Complexity score $C \in [0, 1]$
M5: Defender Agent	Formulates corroborative queries with stance = supporting.	Supporting Evidence Pool
M6: Prosecutor Agent	Formulates adversarial counter-queries targeting anchor disputes & refutations.	Opposing Evidence + Challenges
M7: Multi-Turn Debate	2-round evidentiary exchange; fast-path consensus early exit at $S \geq 0.70$.	Jaccard Lift J(D, P)
M8: Quality Firewall	Intrinsic quality scoring (relevance, credibility, freshness) & clustering.	Kish Sample Size (n_{eff})
M9: Cross-Encoder NLI	DeBERTa-v3 evaluates winning cluster representatives for entailment/refutation.	Posterior: P_{sup}, P_{con}, P_{neu}
M10: Rollup & Calibration	Bayesian posterior aggregation across atomic claims into response-level index.	Global Hallucination Index H
M11: Verification API	FastAPI endpoints (/api/v1/verify) with SQLite audit persistence.	Standardized JSON Schema
M12: Dashboard UI	SPA dashboard displaying supportive claims, opposing claims, & internet URLs.	Interactive Web Interface

2. Mathematical Foundations & Verification Protocol

The verification pipeline implements mathematically grounded algorithms to score evidence quality, aggregate adversarial perspectives, eliminate redundancy, and calibrate Bayesian posterior probabilities.

A. Linguistic Qualifier Extraction & Complexity Scoring (Module 3 & 4)

Every proposition P is parsed into an atomic claim tuple $C = (s, p, o, T, N, S, Q)$, where T represents temporal anchors (years, dates), N denotes numerical quantities, S represents spatial entities, and Q denotes factual qualifiers. The claim complexity $C(\text{claim})$ dictates retrieval depth:

```
Complexity Score: C(claim) = min(1.0, 0.25·(#words/20) + 0.30·|T| + 0.25·|N| + 0.20·|S| + 0.15·|Q|)
```

B. Adversarial Debate Protocol & Jaccard Complementarity (Module 5, 6, & 7)

To prevent one-sided confirmation bias, evidence is retrieved through dual adversarial agents. The Defender gathers corroborative evidence D, while the Prosecutor identifies contested anchors and gathers refutation evidence P. The evidentiary diversity is governed by the Jaccard Complementarity metric $J(D, P)$:

```
Jaccard Complementarity: J(D, P) = 1.0 - (|D ∩ P| / |D ∪ P|) ∈ [0.0, 1.0]
```

When Round 1 yields high-confidence evidence (composite score $S_{\text{composite}} \geq 0.70$), an early consensus termination triggers, reducing latency by 50%.

C. Multi-Signal Quality Firewall & Kish Effective Sample Size (Module 8)

Each evidence item e_i receives an intrinsic composite score combining relevance, domain authority/credibility, and temporal freshness:

```
Composite Score: S_composite(e_i) = 0.50·S_rel + 0.30·S_cred + 0.20·S_fresh (Gated at S ≥ 0.20)
```

To prevent duplicate citations (e.g., 5 identical news reposts) from artificially inflating confidence, we compute **Kish's Effective Sample Size** (n_{eff}) over cluster reliability weights w_i:

```
Kish Effective Sample Size: n_eff = ( ∑_{i=1}^M w_i )^2 / ∑_{i=1}^M w_i^2
```

D. DeBERTa-v3 Cross-Encoder Adjudication & Global Hallucination Index (Module 9 & 10)

Downstream NLI cross-attention is executed on cluster representatives using DeBERTa-v3, evaluating pairs (*Premise* = Evidence Snippet, *Hypothesis* = Atomic Claim) to compute softmax probabilities $[p_{\text{ent}}, p_{\text{con}}, p_{\text{neu}}]$. Individual predictions are combined using weighted Bayesian log-odds:

```
Log-Odds Combination: L_post = L_prior + ∑_{k=1}^K w_k · ln( P(e_k | H_sup) / P(e_k | H_con) )
```

Atomic claim verdicts follow a strict tri-class calibrated taxonomy: **SUPPORTED** ($P_{\text{sup}} \geq 0.55$ and $P_{\text{sup}} > 1.25 \cdot P_{\text{con}}$), **REFUTED** ($P_{\text{con}} \geq 0.55$ and $P_{\text{con}} > 1.25 \cdot P_{\text{sup}}$), or **NOT_ENOUGH_INFO**. Across all N atomic claims in a passage, the **Global Hallucination Index** $H \in [0.0, 1.0]$ is calculated:

```
Global Hallucination Index: H = ( ∑_{j=1}^N (1.0 - P_sup(c_j)) · C(c_j) ) / ( ∑_{j=1}^N C(c_j) )
```

3. Federated Retrieval, Empirical Validation & UI Dashboard

To ensure zero API costs and real-world domain coverage, the system implements a dual-track Federated Evidence Retriever that escalates seamlessly between offline benchmark corpora and live internet search.

A. Federated Search Architecture & Entity Grounding Escalation

1. **Local Benchmark Corpora (BM25):** Indexes peer-reviewed biomedical abstracts (SciFact / PubMed) and general knowledge encyclopedic articles (FEVER / Wikipedia).
2. **Live Web Ingestion (DuckDuckGo + Wikimedia REST API):** Retrieves contemporary online documentation with full URL and snippet extraction.
3. **Entity Grounding Firewall:** When a query contains distinct alphanumeric identifiers (e.g. PCSK9, COVID-19, BRCA1), the retriever validates that local benchmark matches actually ground the entity. If missing, it immediately escalates to live web retrieval.

B. Empirical Benchmark Results & Case Studies

Test Statement / Proposition	Retrieval Track	Adjudicated Verdict	H-Index	Sources & Key Findings
Cardiovascular Fact: *'Aspirin reduces cardiovascular mortality and lowers myocardial infarction risk.'*	SciFact / PubMed (BM25)	**SUPPORTED** (95% Conf)	**0.05** (Highly Factual)	• Clinical trials (PMID: 15282478) • Entailment score: 94.2%
Genetic Hallucination: *'Loss-of-function mutations in PCSK9 cause severe hypercholesterolemia.'*	Live Web (DuckDuckGo + Wikipedia)	**REFUTED** (91% Conf)	**0.95** (Severe Hallucination)	• 7 live sources (MedlinePlus, PMC5729040) • Loss-of-function *lowers* LDL & protects heart
Mixed Veracity Claim: *'High-dose Vitamin C cures acute viral hepatitis and completely prevents influenza.'*	Live Web / Hybrid	**REFUTED** (89% Conf)	**0.92** (Severe Hallucination)	• NIH / Cochrane reviews dispute cure claims • Prosecutor identified predicate refutation

C. Interactive Web UI Dashboard & Review Demonstration

Module 12 provides a single-page application (SPA) dashboard served via FastAPI (http://localhost:8000). In alignment with project review requirements, the output is structured into three clear visual pillars:

1. ■ **Supportive Claims & Evidence:** Displays verified evidence excerpts with green indicator cards, source document titles, and NLI entailment confidence percentages.
2. ■ **Opposing Claims & Evidence:** Highlights contradicting passages in red indicator cards with refutation confidence percentages and structured challenge tags.
3. ■ **Internet Sources & References:** Provides direct clickable URLs to original sources (e.g. medlineplus.gov, pmc.ncbi.nlm.nih.gov, springer.com, en.wikipedia.org) and snippet previews.

D. Performance Benchmarks & Key Technical Specifications

Metric / Benchmark	Measured Value	Technical Design Rationale
Automated Test Suite	**8 / 8 Tests Passing (100%)**	Full regression suite across single-claim, multi-claim, live web, and batch verification.
End-to-End Latency	**~9.4s (warm CPU) / <13s (batch)**	Pruned cluster evaluation (top 2 sup + top 2 con) and PyTorch 2-thread cap.
Hardware Overhead	**0 GPU required | < 1.8 GB RAM**	Lightweight CPU inference; bounded chunking prevents Windows paging memory exhaustion.
Operating Cost	**$0.00 (100% Free & Open Source)**	Zero dependencies on OpenAI, Anthropic, or paid search APIs; fully self-contained.